Social Networks and their Analysis

LINK ANALYSIS

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Social Networks and their Analysis

Homophily, Social Influence, and Affiliation

- *Homophily Test*
- *Selection and Social Influence*
- *Affiliation Networks*
- *Schelling Model of Segregation*
- *Positive and Negative Relationships*

Link Analysis

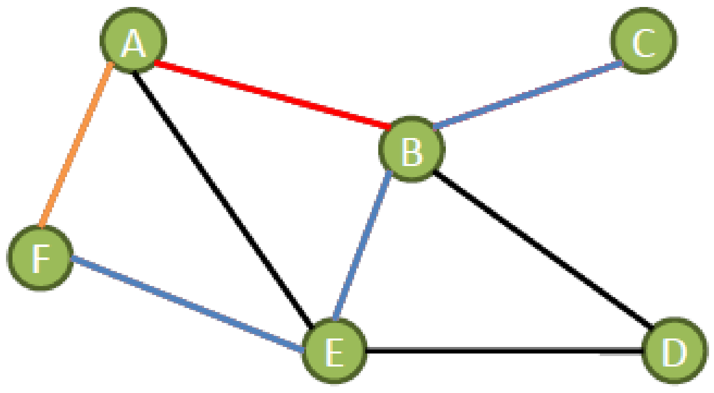
- **A Brief Review of Actor-Level and Network-Level Measures**
- *Hubs and Authorities, the HITS Algorithm*
- *The PageRank Algorithm*
- *PageRank – Revisited*
- *HITS – Revisited*
- *Themed Communities and Their Discovery*

Statistical Measures to Analyze Networks: Basic Concepts

Path: sequence of nodes in which consecutive pairs of (non-repeating) nodes are linked by an edge; the first vertex of a path is called the *start vertex* and the last vertex of the path is called the *end vertex*

Geodesic distance: the geodesic distance between two nodes/vertices is given by the number of edges connecting them in the shortest path; length of the shortest path

Eccentricity of a vertex: greatest geodesic distance between a given vertex *v* and any other in the graph



Path

Eccentricity of vertex C

Greatest shortest path= {(C,B),(B,E),(E,F)}
e(C)=3

$$\epsilon_v = \max_{i \in V(G) \setminus v} d(v,i)$$

$d_{C,A}=2$

Statistical Measures to Analyze Networks

Actor-Level measures:

Measures of Centrality: *centrality* is a general measure of how the position of a vertex is within the overall structure of the graph; it helps identify the key players in the network. The best known are:

- Degree, Valency or Prestige
  - In-degree
  - Out-degree
- Betweenness
- Closeness
- Eigenvector centrality

Network-Level measures: to assess the overall structure of the network

- Diameter/Radius
- Average geodesic distance
- Average degree
- Reciprocity
- Density
- Clustering

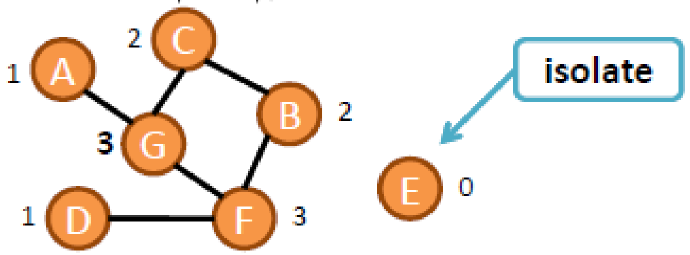
Statistical Measures to Analyze Networks: Actor-Level Measures

Degree, Valency or Prestige

- Number of neighbors (or connections) of a vertex *v*
- It is a measure of the number of individuals in the network that a specific actor can reach or, alternatively, it is a measure of the involvement of the actor in the network

Undirected networks:

$$k_v = |N_v|, 0 < k_v < n$$



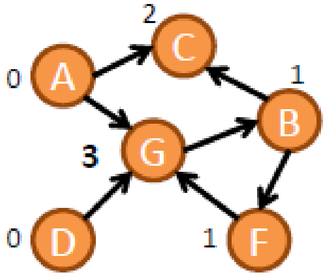
Directed networks:

In-degree	Number of incoming vertices Measure of support	k_v^+
Out-degree	Number of outgoing vertices Measure of influence	k_v^-

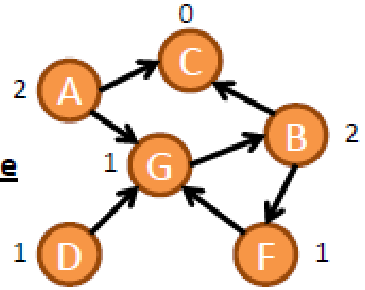
Weighted networks:

$$k_v^w = \sum_{u \in N_v} w_{vu}$$

In-degree



Out-degree



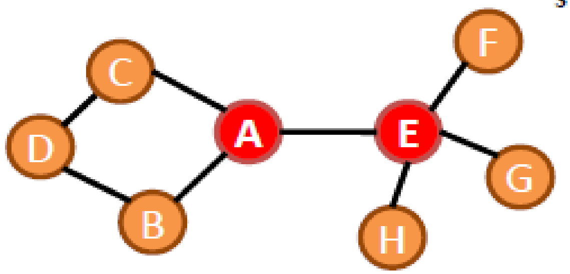
Statistical Measures to Analyze Networks: Actor-Level Measures

Betweenness

- The extent to which a node lies between other nodes in the network
- Vertices with high betweenness occupy critical roles in the network structure, since they usually have a network position that allow them to work as an interface between tightly-knit groups, being "vital" elements in the connection between different regions of the network.
- In the *Social Networks* context, these nodes are known as the **gatekeepers**

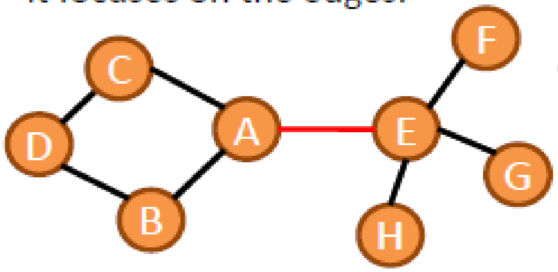
$$b_v = \sum_{s,t\in V(G)\setminus v} \frac{\sigma_{st}(v)}{\sigma_{st}}$$

Total number of shortest paths between vertices *s* and ***t***



Edge betweenness:

The idea is the same but instead of focusing on nodes, it focuses on the edges.



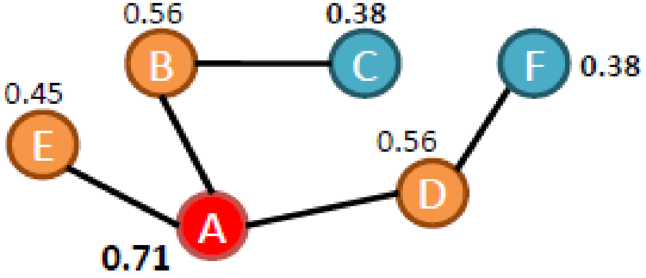
$$b_e = \sum_{u,v\in V(G)} \frac{\sigma_{uv}(e)}{\sigma_{uv}}$$

Statistical Measures to Analyze Networks: Actor-Level Measures

Closeness ~ inverse mean length of all shortest paths

- Mean length of all shortest paths from one node to all other nodes in the network
- Only computed for vertices within the largest component of the network
- Measure of reachability that gives an idea about how long it will take to reach other nodes from a given starting node
- In the *Social Network* context, closeness measures how fast can a given actor reach everyone in the network

$$Cl_v = \frac{n-1}{\sum_{u \in V(G)\setminus v} d(u,v)}$$



Node A has the highest closeness to all other nodes in the network.

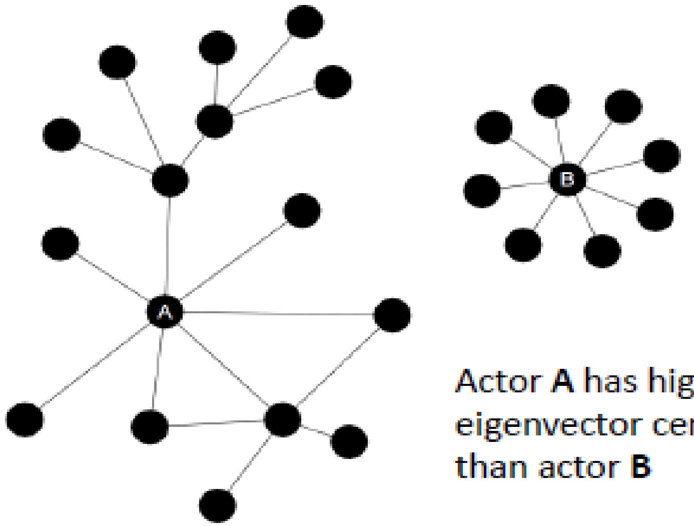
Nodes C and F are the ones with lowest closeness in the network.

Statistical Measures to Analyze Networks: Actor-Level Measures

Eigenvector Centrality

- This metric is based on the idea that the power and status of an actor is recursively defined by the power and status of his/her alters (or neighbors)
- The eigenvector centrality of a node is proportional to the sum of the eigenvector centralities of all its direct neighbors
- It measures how well a given actor is connected to other well-connected actors

Degree VS Eigenvector centrality: **eigenvector centrality** is a more elaborated version of the **degree**, once it assumes that not all connections have the same importance by taking into account not only the quantity, but especially the quality of these connections.



Actor **A** has higher eigenvector centrality than actor **B**

Statistical Measures to Analyze Networks: Actor-Level Measures

For a given network $G = (V, E)$, let $A = (a_{v,t})$ be the adjacency matrix, i.e., $a_{p,t} = 1$ if vertex v is linked to vertex t, and $a_{v,t} = 0$ otherwise. The *relative centrality score* of vertex v can be defined as:

$$x_v = \frac{1}{\lambda} \sum_{t \in M(v)} x_t = \frac{1}{\lambda} \sum_{t \in G} a_{v,t} x_t$$

where $M(v)$ is a set of the neighbors of v and λ is a constant. With a minor rearrangement this can be rewritten in vector notation as the eigenvector equation

$$\lambda \mathbf{x} = \mathbf{A}\mathbf{x}$$

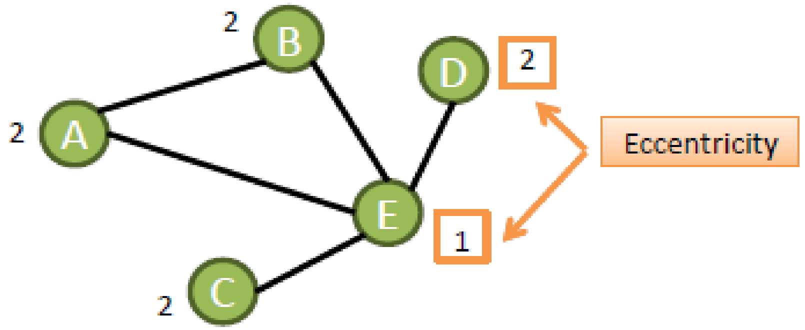
- many different eigenvalues λ for which a non-zero eigenvector solution exists
- the additional requirement that all the entries in the eigenvector be non-negative implies that only the greatest eigenvalue results in the desired centrality measure
- The eigenvector is only defined up to a common factor – we must normalize the eigenvector, e.g., sum over all vertices is 1 or the total number of vertices n.

Statistical Measures to Analyze Networks: Network-Level Measures

Diameter/Radius

Diameter:
- Longest shortest path between any two nodes in the network
- Maximum eccentricity of the set of vertices in the network
- Sparser networks have generally greater diameter

Radius:
- Minimum eccentricity of the set of vertices in the network



Eccentricity of a vertex ~ greatest geodesic distance between a given vertex and any other vertex in the graph

Diameter= 2

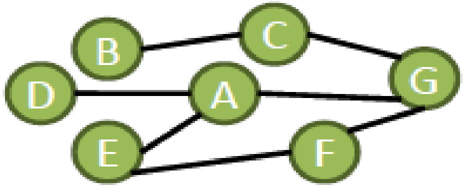
Radius = 1

Statistical Measures to Analyze Networks: Network-Level Measures

Average Geodesic Distance

- The average geodesic distance gives an idea of how far apart nodes will be, on average, in the network
- Measures the efficiency of information flow within the network

$$l = \frac{1}{\frac{1}{2}n(n+1)} \sum_{i \geq j} d_{ij}$$



Normalization includes the diagonal

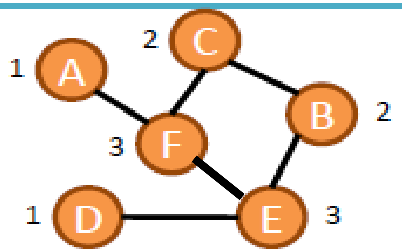
(without the diagonal: $\frac{1}{2}n(n-1)$)

$$l = \frac{3+2+1+1+2+1+1+4+4+3+2+3+3+2+1+2+3+2}{\frac{1}{2}7(7+1)} = \frac{40}{28} \approx 1.43$$

Average Degree

- Measure of the overall connectivity of the network
- Computed as the mean of the degrees of all network's vertices

$$\overline{k} = \frac{1}{n}\sum_{i=1}^{n} k_i$$



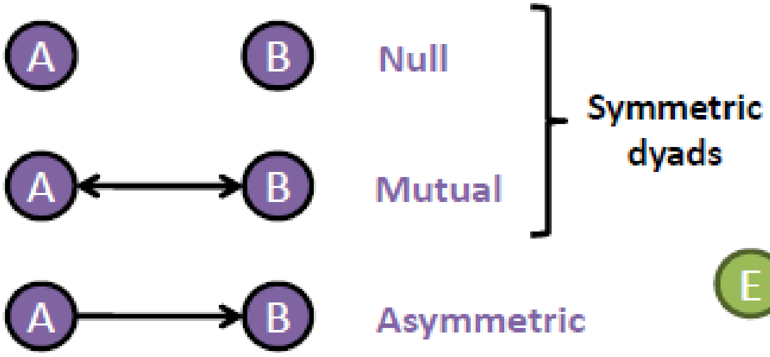
$$\overline{k} = \frac{1+3+2+2+3+1}{6} = \frac{12}{6} = 2$$

Statistical Measures to Analyze Networks: Network-Level Measures

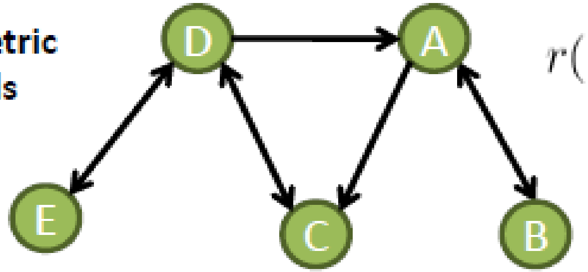
Reciprocity

- Measures the tendency of the pairs of vertices to form reciprocal connections between each other
- Specific metric for directed networks
- *The reciprocity index* $r(D)$ is computed as the proportion of symmetric dyads in given digraph and its value represents the probability that two vertices share the same type of connection.

Classes of isomorphism for dyads



$$r(D) = \frac{s(D)}{C_2^n} = \frac{mut(D) + null(D)}{C_2^n}, 0 < r < 1$$



$$r(D) = \frac{3+5}{C_2^5} = \frac{8}{10} = 0.8$$

$$C_2^n = \binom{n}{2} = \frac{n(n-1)}{2}$$

Statistical Measures to Analyze Networks: Network-Level Measures

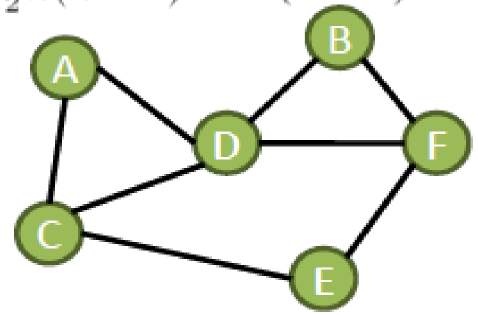
Density

- The *density* indicates the level of connectedness in a network, with high values being associated to **dense networks** and low values associated to **sparse networks**
- Density is simply the proportion of edges/arcs (m) in the graph/digraph relative to the maximum possible number of edges/arcs (m_{max})
- A perfectly connected network is called a ***clique***, or ***complete graph***, and has a maximum density of 1

Undirected networks:

$$\rho(G) = \frac{m}{m_{max}} = \frac{m}{\frac{1}{2}n(n-1)} = \frac{2m}{n(n-1)},\ 0 < \rho < 1$$

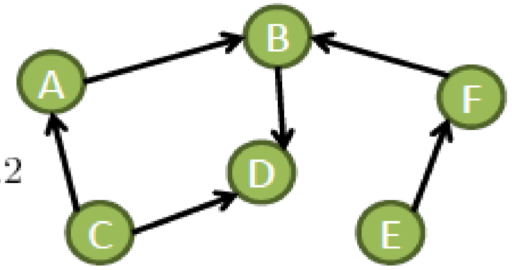
$$\rho(G) = \frac{2\times 8}{6(6-1)} \approx 0.53$$



Directed networks:

$$\rho(D) = \frac{m}{m_{max}} = \frac{m}{n(n-1)}$$

$$\rho(D) = \frac{6}{6(6-1)} = 0.2$$

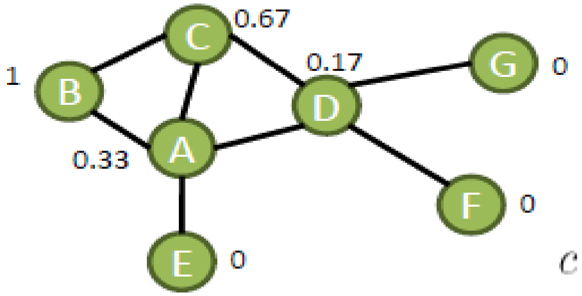


Statistical Measures to Analyze Networks: Network-Level Measures

Transitivity or Clustering

- *Transitivity*, or *clustering*, is a property that considers the density/cohesion of a node's neighborhood, in its **local version**; or the density of triangles in a network, in its **global version**.
- This property is quantified by computing a ***clustering coefficient***, that can be local (computed for each node) or global (computed for the whole network)
  - ✓ **Local clustering coefficient:** fraction of pairs of vertices, that are neighbors of a given vertex *v*, that are connected to each other by edges.
  - ✓ **Global clustering coefficient:** average of all local coefficients
- Clustering is useful once it indicates the presence of sub-communities in the network

Local clustering coefficients



$$c_i = \frac{2|e_{jk}|}{k_i(k_i-1)} : v_j, v_k \in N_i, e_{jk} \in E$$

Local

Global clustering coefficient

$$c = \frac{1+0.67+0.33+0.17}{7} = 0.31$$

$$c = \frac{1}{n}\sum_v c_v$$

Global

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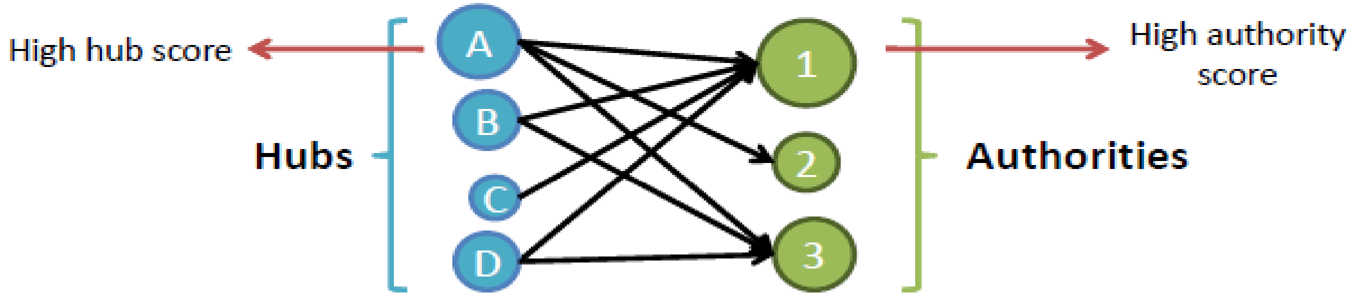
Link Analysis: Hubs and Authorities

Authorities

- Authorities are web pages cited by many different *hubs*
- Good *authorative pages* are reliable sources of information about a given topic
- Using the network parlance, *authorities* are nodes receiving many inward links
- The relevance of an authority is “measured” by the number of inward links

Hubs

- Hub can be understood as a web page that points to many other web pages or, in other words, as a compilation of web pages that address a specific topic
- Using the network parlance, *hub* is a node with many outward links
- A good hub is a site that points to good *authorative sites*



Link Analysis: Hubs and Authorities

HITS algorithm

- **HITS** (*Hypertext Induced Topic Selection*) is a link analysis algorithm developed by Kleinberg (1999)
- HITS computes and returns 2 scores, for each node *v* in the network: the **authority score** *auth(v)* and the **hub score** *hub(v)*
- These scores provide information about the potential of each node to be an *authority* or a *hub*, thus indicating how valuable is the information carried by it

Link Analysis: Hubs and Authorities

HITS algorithm

HITS is an iterative algorithm based on the repeated update of two basic rules:

- **Authority Update Rule:** for each page *p* (or node *v*), update ***auth(p)*** to be the sum of the hub scores of all pages that point to it. A high authority score is assigned to *p* if it is linked to pages that are recognized as important hubs of information.

- **Hub Update Rule:** for each page *p* (or node *v*), update ***hub(p)*** to be the sum of the authority scores of all pages that it points to. A high hub score is assigned to *p* if it links to nodes that are considered to be authorities in a given topic.

Link Analysis: Hubs and Authorities

HITS – algorithm:

- Given a query (topic) $\boldsymbol{Q}$, collect the root set of pages $\boldsymbol{S} = \{\boldsymbol{s_1}, \boldsymbol{s_2}, \boldsymbol{s_3}, \dots, \boldsymbol{s_n}\}$
- Expand the set $\boldsymbol{S}$ to set $\boldsymbol{T} = \boldsymbol{S} \cup \{\boldsymbol{d} | \boldsymbol{s} \to \boldsymbol{d} \vee \boldsymbol{d} \to \boldsymbol{s};\ \boldsymbol{s} \in \boldsymbol{S}\}$

Initialization: set $\boldsymbol{auth(p) = 1}$, $\boldsymbol{hub(p) = 1}\ \forall \boldsymbol{p} \in \boldsymbol{T}$ and the number of updates $\boldsymbol{k}$

1. Run the Authority Update Rule
2. Run the Hub Update Rule
3. Perform $\boldsymbol{k}$ Authority / Hub updates (according to previous rules)

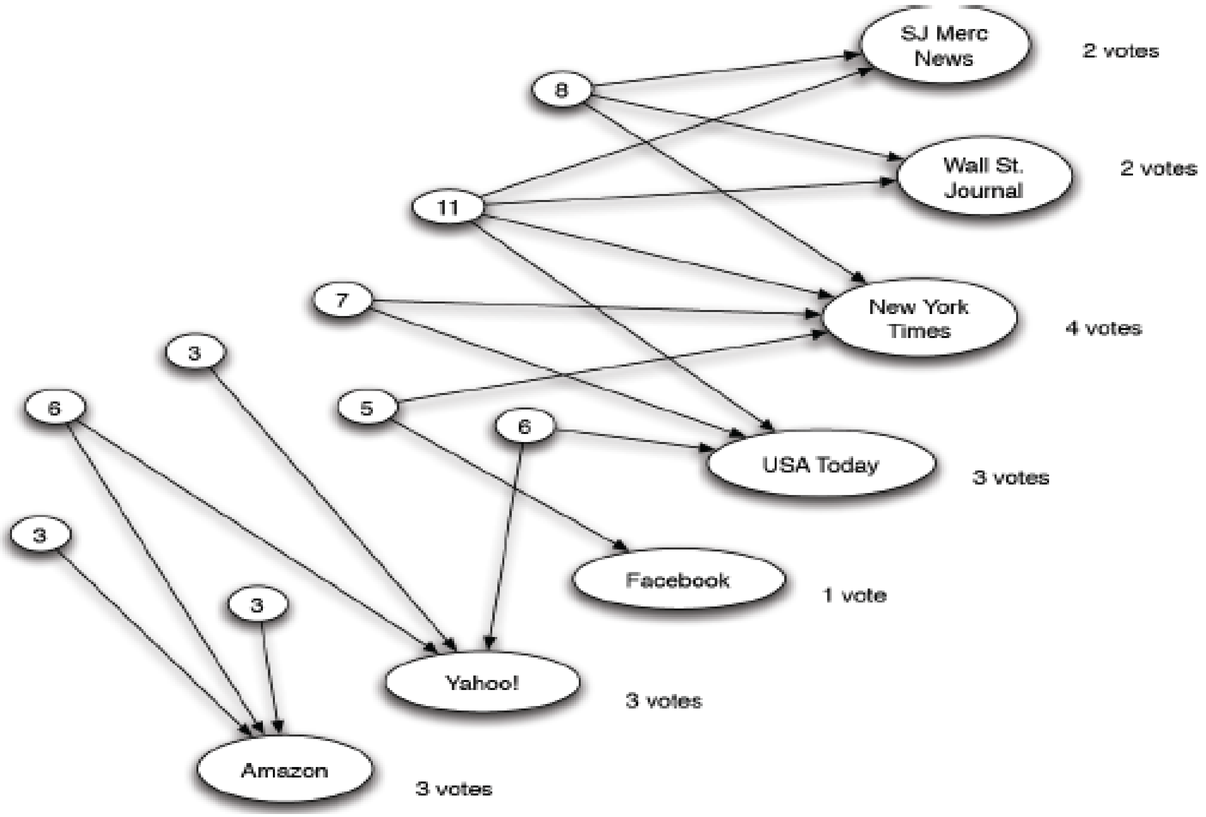
Finally, normalize the values $\boldsymbol{auth(p)}$ and $\boldsymbol{hub(p)}$ due to their tendency to grow large.

For normalization, divide authority / hub scores by the sum of all authority / hub scores.

Link Analysis: Hubs and Authorities

HITS algorithm: toy example
k=1

(Easley and Kleinberg, 2010)



Query *Q*:"newspapers"

k=1

→<u>Hubs</u> are represented by small circles

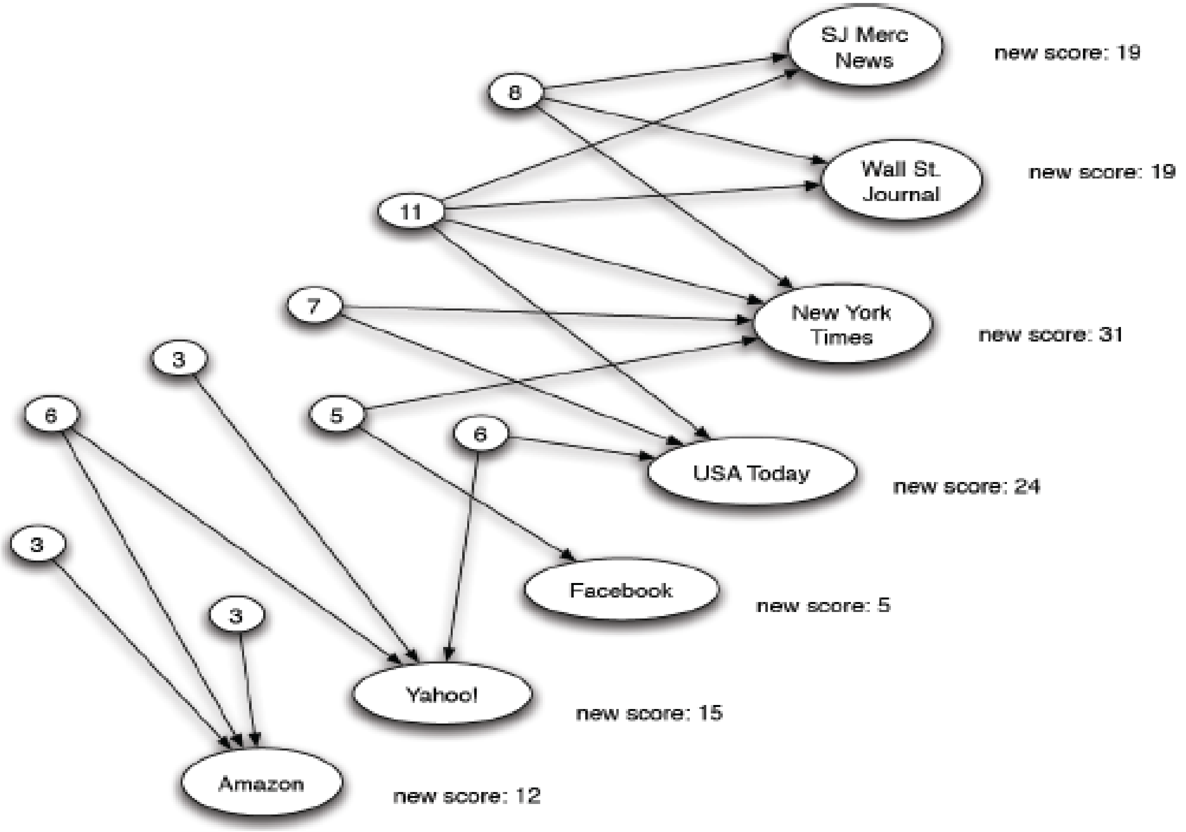
→<u>Authorities</u> are represented by elongated circles

Each circle has assigned the *hub score* and the *authority score* of each page after the first iteration (k=1) of HITS algorithm

Link Analysis: Hubs and Authorities

HITS algorithm: toy example
k=2

(Easley and Kleinberg, 2010)



Query *Q*:“newspapers”

→Hubs are represented by small circles
→Authorities are represented by elongated circles

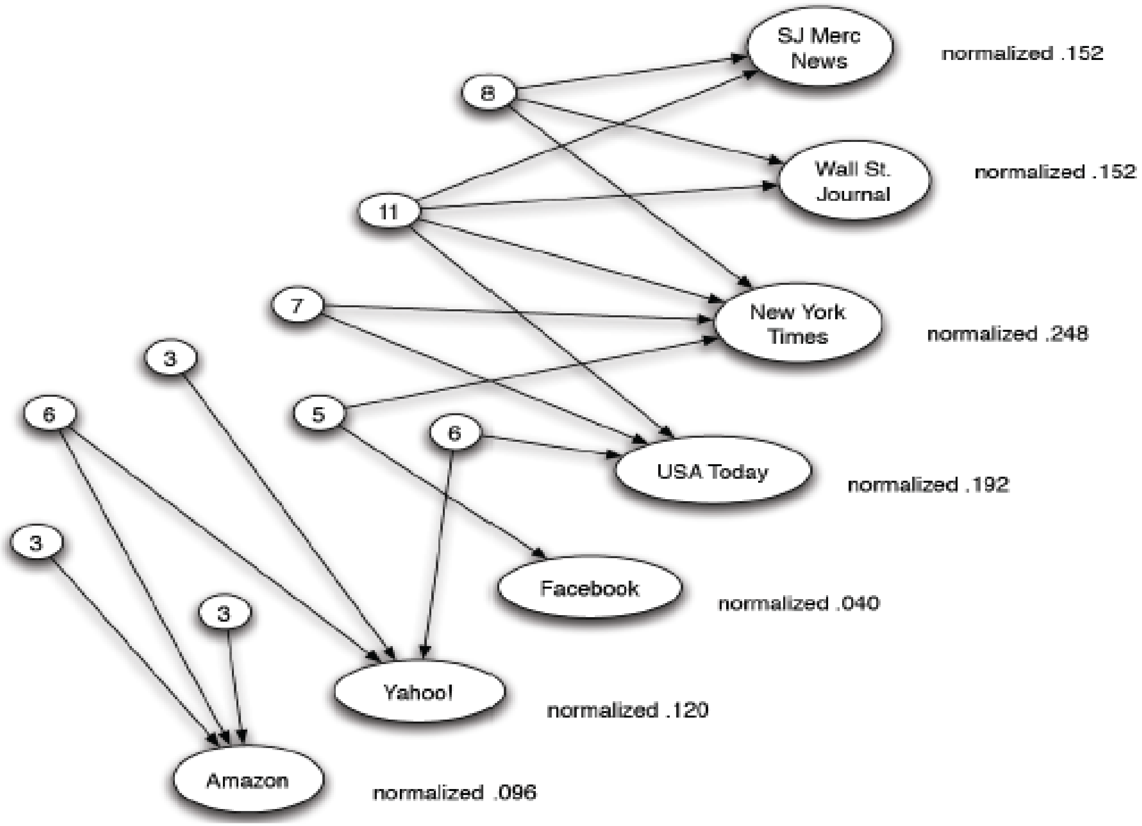
Second iteration of the algorithm (k=2).

E.g. Amazon is cited by 3 hubs, thus the new score of this page will be, according to the ***authority update rule***, the sum of the values of all hubs that point to it (new score=3+3+6=12)

Link Analysis: Hubs and Authorities

HITS algorithm: toy example k=3

(Easley and Kleinberg, 2010)



Query *Q*:"newspapers"

→<u>Hubs</u> are represented by small circles

→<u>Authorities</u> are represented by elongated circles

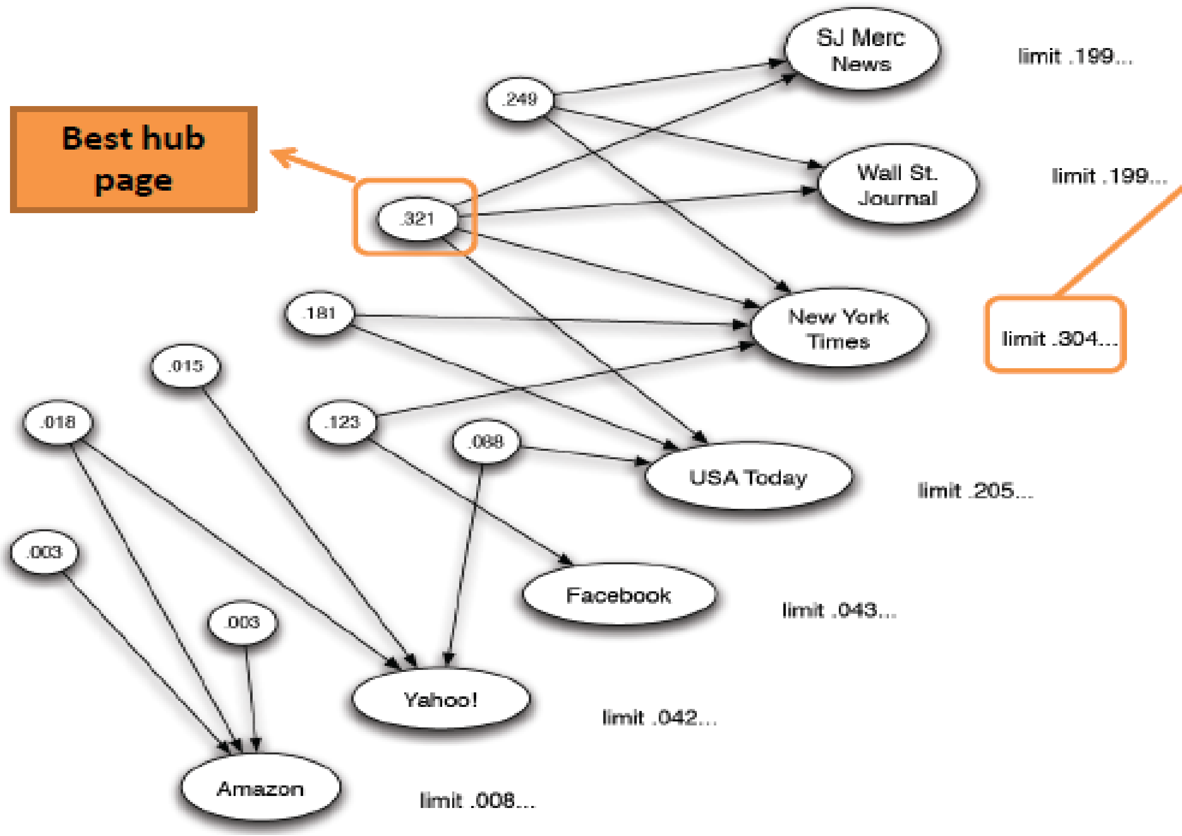
Third and last iteration of the algorithm (k=3). Normalization of the returned authority scores.

$\sum auth(p)$= 19+19+31+24+5+15+12= 125
Norm.auth(Amazon)= 12/125= 0.096
Norm.auth(Yahoo!)=15/125= 0.120
(...)

Link Analysis: Hubs and Authorities

HITS algorithm: toy example
Convergence of normalized scores

(Easley and Kleinberg, 2010)



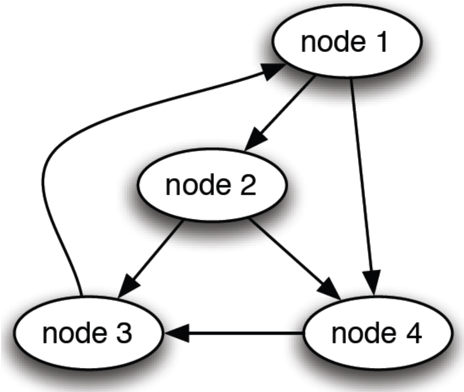
Best authority page:
"New York Times"

Query *Q*:"newspapers"

→ If we run HITS algorithm for very large values of *k* (infinite *k*), the normalized scores for hubs and authorities will converge to limits.

→ These limits are represented in the network.

$$\begin{bmatrix} 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$



Link Analysis:
Hubs and Authorities

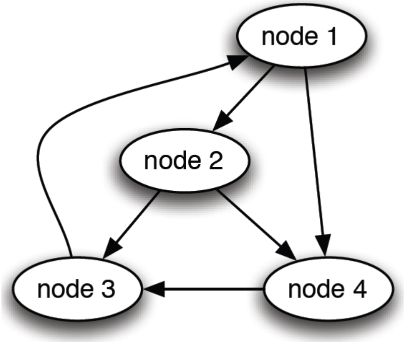
Spectral Analysis of HITS

- Spectrum of a matrix is the set of its eigenvalues
- Adjacency Matrix - $n \times n$ matrix M with:
  - $M_{ij} = 1$ if there is a link from node i to node j,
  - $M_{ij} = 0$ otherwise
- Hub/Authority Scores
  - hub (h) and authority (a) vectors $n \times 1$
- write the *Authority Update* and *Hub Update Rules* as matrix-vector multiplications

$$h_i := M_{i1}a_1 + M_{i2}a_2 + \cdots + M_{in}a_n$$
$$a_i := M_{1i}h_1 + \cdots + M_{ni}h_n$$

$$h := Ma$$
$$a := M^T h$$

Link Analysis: Hubs and Authorities



$$\begin{bmatrix} 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \\ 1 \\ 2 \end{bmatrix}$$

$h_0 = [\ 1\ 1\ 1\ 1\ ]^T$

$= a_1$

$= h_1$

Spectral Analysis of HITS

- Adjacency Matrices
  - $n \times n$ matrix M: $M_{ij} = 1$ if link $i \to j$, $M_{ij} = 0$ otherwise
- Hub/Authority Scores
  - hub (h) and authority (a) vectors $n \times 1$
- write the *Authority Update* and *Hub Update Rules* as matrix-vector multiplications

$$h_i := M_{i1}a_1 + M_{i2}a_2 + \cdots + M_{in}a_n$$
$$a_i := M_{1i}h_1 + \cdots + M_{ni}h_n$$

$$\Rightarrow \quad h := Ma$$
$$a := M^T h$$

Link Analysis: Hubs and Authorities

k step Hub-Authority Computation

- Initial vectors of authority and hub scores $a^{\langle 0 \rangle}$ and $h^{\langle 0 \rangle}$
- After step 1

$$a^{\langle 1 \rangle} = M^T h^{\langle 0 \rangle}$$
$$h^{\langle 1 \rangle} = M a^{\langle 1 \rangle} = M M^T h^{\langle 0 \rangle}$$

- After step 2

$$a^{\langle 2 \rangle} = M^T h^{\langle 1 \rangle} = M^T M M^T h^{\langle 0 \rangle}$$
$$h^{\langle 2 \rangle} = M a^{\langle 2 \rangle} = M M^T M M^T h^{\langle 0 \rangle} = (M M^T)^2 h^{\langle 0 \rangle}$$

- After step k

$$a^{\langle k \rangle} = (M^T M)^{k-1} M^T h^{\langle 0 \rangle}$$
$$h^{\langle k \rangle} = (M M^T)^k h^{\langle 0 \rangle}$$

Link Analysis: Hubs and Authorities

- After step k

$$a^{\langle k\rangle} = (M^T M)^{k-1} M^T h^{\langle 0\rangle}$$
$$h^{\langle k\rangle} = (M M^T)^{k} h^{\langle 0\rangle}$$

- Magnitude of the hub and authority values tends to grow with each update
- They will only converge when we take normalization into account
- There are (normalizing) constants c and d s.t.: $\frac{h^{\langle k\rangle}}{c^k}$ and $\frac{a^{\langle k\rangle}}{d^k}$ converge to limits as $k \to \infty$
- Taking the recursive formula: $\frac{h^{\langle k\rangle}}{c^k} = \frac{(MM^T)^k h^{\langle 0\rangle}}{c^k}$
- $h^{\langle k\rangle}$ converges to the limit $h^{\langle * \rangle}$ if the direction does not change when multiplied with MM^T (but magnitude may change by a factor c)

$$(MM^T)h^{\langle * \rangle} = c h^{\langle * \rangle}$$

Link Analysis: Hubs and Authorities

- *Q:* When a vector v does not change its direction if multiplied by a given matrix X?
- *A:* When v is an eigenvector of X
  - $Xv = \lambda v$
  - A solution of: $\det(X - \lambda I) = 0$
- It follows that $h^{\langle * \rangle}$ has to be an eigenvector of MM^T
- We have to prove that the direction of $h^{\langle k \rangle}$ (normalized: $\frac{h^{\langle k \rangle}}{c^k}$) converges to an eigenvector of MM^T
- MM^T is symmetric => has n eigenvectors $v_1, v_2, \dots, v_n$ with the corresponding eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$ (assume w.l.g. that: $|\lambda_1| \geq |\lambda_2| \geq \cdots \geq |\lambda_n|$)

$$h^{\langle k \rangle} = (MM^T)^k h^{\langle 0 \rangle} = (MM^T)^k (q_1 v_1 + \cdots + q_n v_n) =$$

$$= q_1 (MM^T)^k v_1 + \cdots + q_n (MM^T)^k v_n = q_1 \lambda_1^k v_1 + \cdots + q_n \lambda_n^k v_n$$

Link Analysis: Hubs and Authorities

How to find an eigenvector?

1. Find the eigenvalues of the given matrix X by solving the equation $\det(X - \lambda I) = 0$. Denote each eigenvalue of X as $\lambda_1, \lambda_2, \ldots$
2. Substitute the values in the equation $Xv = \lambda_1 v$ or $(X - \lambda_1 I)v = 0$
3. Calculate the eigenvector v, which is associated with the eigenvalue λ
4. Repeat the steps to find the eigenvectors for the remaining eigenvalues.

For real symmetric matrices, all eigenvalues are real and eigenvectors corresponding to distinct eigenvalues are orthogonal.

Link Analysis: Hubs and Authorities

- MM^T is symmetric => has n eigenvectors $v_1, v_2, \dots, v_n$ with corresponding eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$ (assume w.l.o.g. that: $|\lambda_1| \geq |\lambda_2| \geq \cdots \geq |\lambda_n|$)

$$h^{\langle k \rangle} = q_1 \lambda_1^k v_1 + \cdots + q_n \lambda_n^k v_n$$

- Assume $|\lambda_1| > |\lambda_2| \geq \cdots \geq |\lambda_n|$

$$\frac{h^{\langle k \rangle}}{\lambda_1^k} = q_1 v_1 + q_2 \left(\frac{\lambda_2}{\lambda_1}\right)^k v_2 \cdots + q_n \left(\frac{\lambda_n}{\lambda_1}\right)^k v_n$$

- For $k \to \infty$, every term except the first goes to 0, hence $\frac{h^{\langle k \rangle}}{\lambda_1^k}$ converges to $q_1 v_1$
- Remaining to prove (not to be shown here)
  - Convergence regardless of initial vector $h^{\langle 0 \rangle}$
  - Relax the assumption $|\lambda_1| > |\lambda_2|$

Link Analysis: Hubs and Authorities

- Endorsement in HITS
- Links denote (collective) endorsement
- Multiple roles in the network:
  - Hubs: pages that play a powerful endorsement role without themselves being heavily endorsed
  - Authorities: pages being heavily endorsed
- Why separate Hubs from Authorities?
  - Competing firms will not link to each other
  - Cannot be viewed as directly endorsing each other
- In **PageRank** endorsement passes directly from one prominent page to another
- A page is important if it is cited by other important pages
  - dominant mode of endorsement in academic or governmental pages, among bloggers, among personal pages, or in scientific literature (pdf's)

Social Networks and their Analysis

Homophily, Social Influence, and Affiliation

- *Homophily Test*
- *Selection and Social Influence*
- *Affiliation Networks*
- *Schelling Model of Segregation*
- *Positive and Negative Relationships*

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- *A Brief Review of Actor-Level and Network-Level Measures*
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Link Analysis: PageRank

PageRank algorithm

- **PageRank** (Brin and Page, 1998) is a link analysis algorithm, which is in the basis of Google's search technology, and it is built upon the concept of eigenvector centrality
- The idea of the algorithm is that information on Web can be ranked according to link popularity: the more web pages are linked to a given web page the more popular that web page is
- In this process of weighting web pages, not only the number of links (degree of a node) is important, but also the importance of the web pages linking to them.

Link Analysis: PageRank

PageRank algorithm

- **Initialization:** in a network of ***n*** nodes, assign a PageRank value of ***1/n*** to each node, and choose the number of iterations ***k***

1. Update the values of each node's PageRank by sequentially applying the following rule:

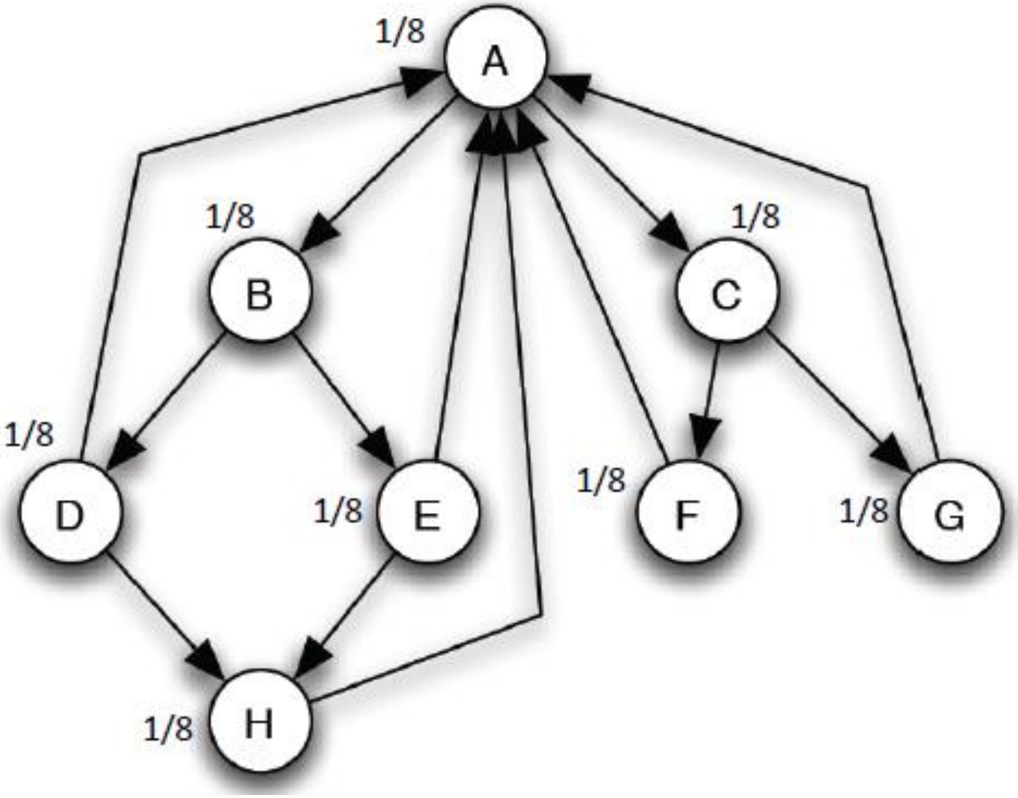
> **Basic PageRank Update Rule:** divide the actual PageRank value of page *p* (or node *v*) by the number of its outgoing links and pass these equal shares to the pages it points to. The update of a node‘s PageRank value is performed by summing the shares it receives in each iteration.

2. Apply this rule until the ***k-th*** iteration.

Link Analysis: PageRank

PageRank algorithm: toy example
Initialization

(Easley and Kleinberg, 2010)



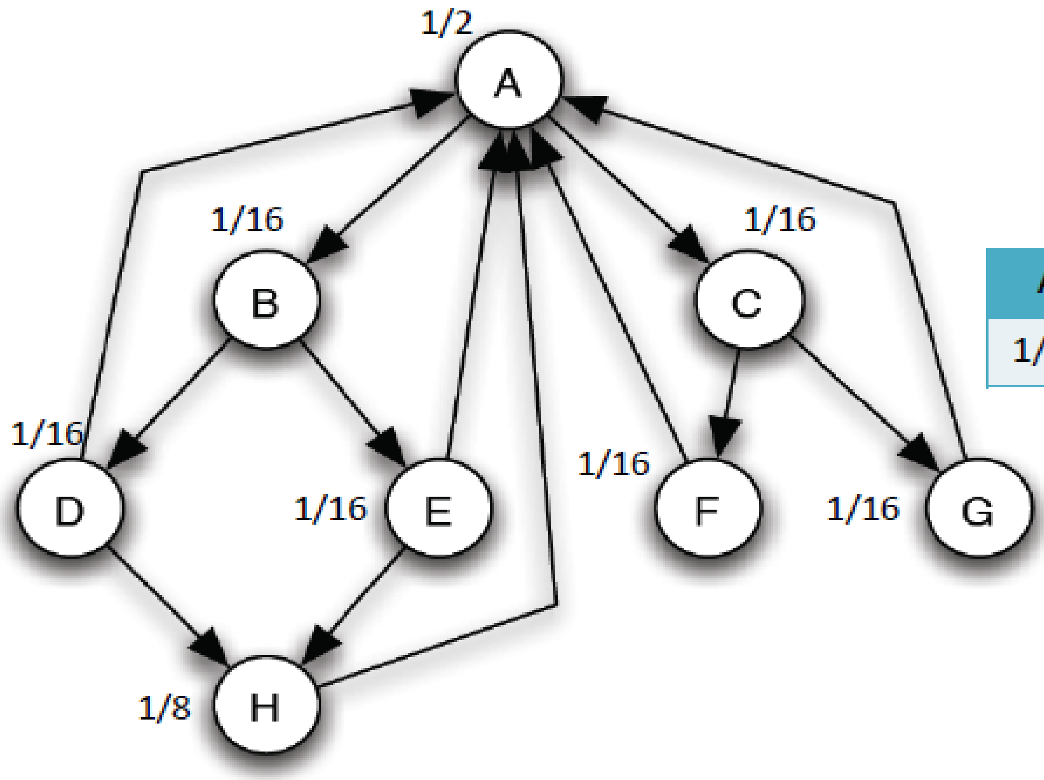
Example: Network comprised of 8 Web pages (n=8)

→ At the <u>initialization step</u>, all pages are assigned a PageRank (PR, from now on) value of 1/8

Link Analysis: PageRank

PageRank algorithm: toy example
k=1

(Easley and Kleinberg, 2010)



→ At the end of the <u>first iteration of the algorithm</u> we obtain new PR values by applying the *Basic PageRank Update Rule*

→ To apply the rule, first is necessary to compute the **shares** of all nodes

A	B	C	D	E	F	G	H
1/16	1/16	1/16	1/16	1/16	1/8	1/8	1/8

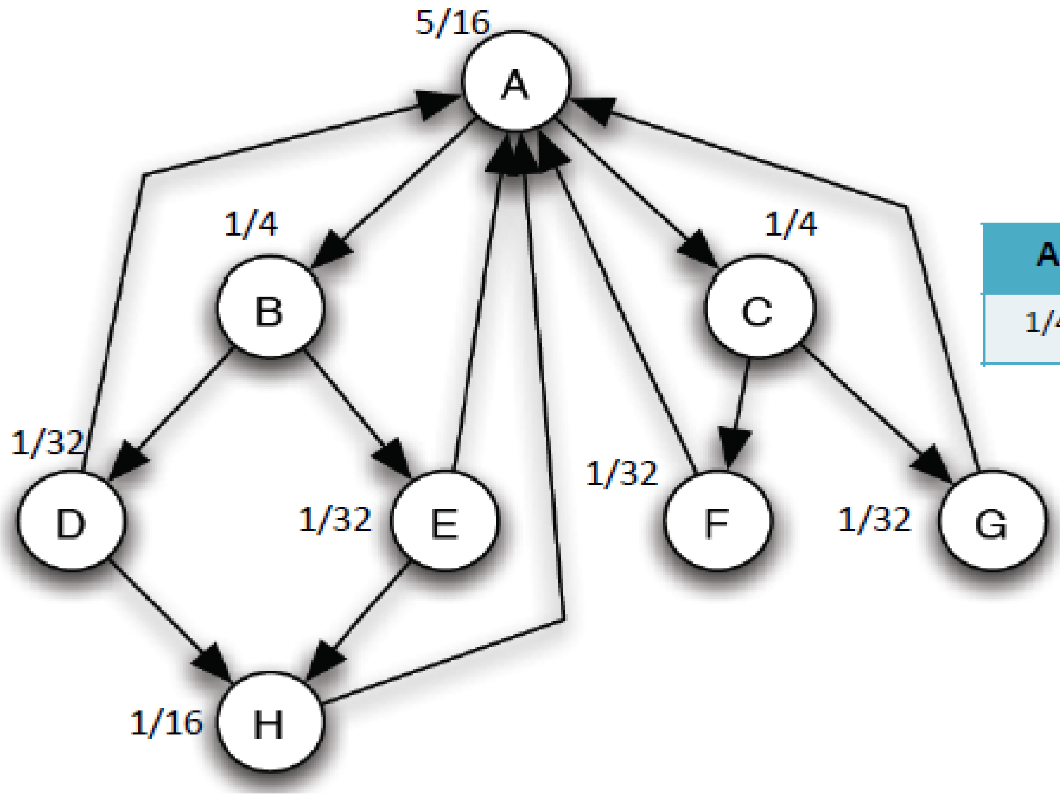
→ Then, for each node we sum all shares the node receives; the result of this sum will be its new PR value

e.g.PR(A)=(1/16)+(1/16)+(1/8)+(1/8)+(1/8)=4/8 =1/2

Link Analysis: PageRank

PageRank algorithm: toy example
k=2

(Easley and Kleinberg, 2010)



→ The rule is applied iteratively until the convergence of PageRank values, or until the *k-th* iteration

→ **Shares of nodes for k=2:**

A	B	C	D	E	F	G	H
1/4	1/32	1/32	1/32	1/32	1/16	1/16	1/8

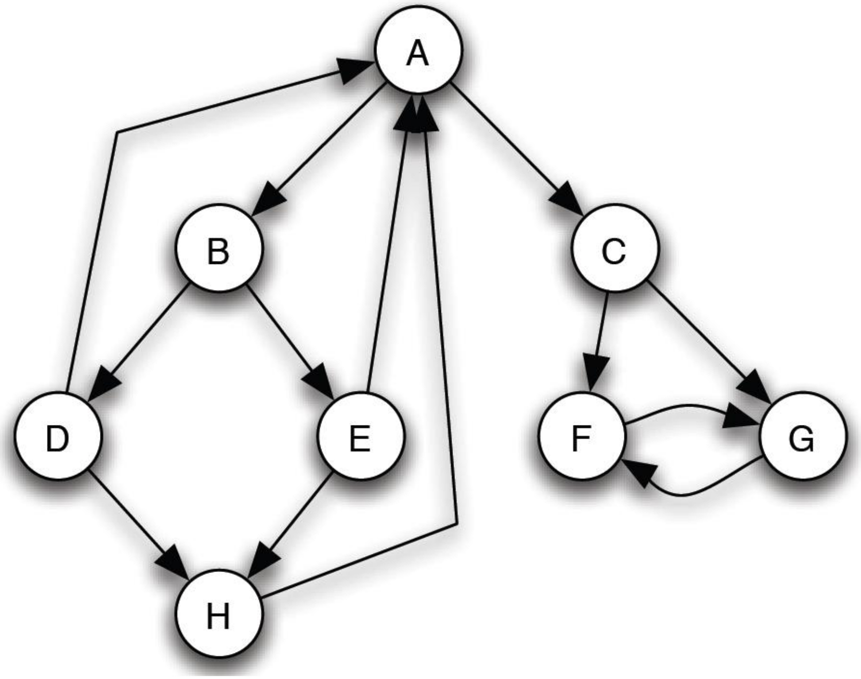
e.g.PR(A)=(1/32)+(1/32)+(1/8)+(1/16)+(1/16)= 5/16

Link Analysis: PageRank

Intuition in PageRank

- PageRank is a kind of "fluid":
  - It circulates through the network
  - Is passing from node to node across edges
  - Is pooling at the nodes that are the most important
- Total PageRank in the network remains constant
  - Why? Each page takes its PageRank, divides it up, and passes it along links
  - PageRank is never created nor destroyed, just moved around from one node to another
- No need to normalize PageRank of nodes to prevent them from growing
  - Unlike HITS

Link Analysis: PageRank



Problems with the Basic Definition of PageRank:

- In many networks, the “wrong” nodes can end up with **all** the PageRank
  - F and G point to each other rather to A
  - PageRank that flows from C to F and G can never circulate back: for large k, we have $\frac{1}{2}$ for each of F and G, and 0 for all other
  - **“slow leak”**: small sets of nodes that can be reached from the rest of the graph, but have no paths back
  - *Solution:* send a small fraction of PageRank to all nodes

Link Analysis: PageRank

Scaling the definition of PageRank

- Pick a scaling factor s between 0 and 1
- Replace the Basic PageRank Update Rule with the following
  Scaled PageRank Update Rule:
  - First apply the Basic PageRank Update Rule
  - Scale down all PageRank values by a factor of s
    - total PageRank in the network has shrunk from 1 to s
  - Divide the residual $1 - s$ units of PageRank equally over all nodes, giving $(1 - s)/n$ to each
- Why does it work?
  - "water cycle" that evaporates $1 - s$ units of PageRank in each step and rains it down uniformly across all nodes

Link Analysis: PageRank

The Limit of the Scaled PageRank Update Rule:

- Repeated application of the Scaled PageRank Update Rule converges to a set of limiting PageRank values as the number of updates k goes to infinity.
- Unique equilibrium
  - But values depend on our choice of the scaling factor s
- The version of PageRank used in practice, with a scaling factor s between 0.8 and 0.9 (scaling factor ~ damping factor)

Link Analysis: PageRank

Spectral Analysis of PageRank

- How PageRank can be analyzed using matrix/vector multiplication and eigenvectors?
- Start with Basic PageRank Update Rule and then move on to the scaled version
- Approach similar to HITS, but there is no need for normalizing
- The "flow" of PageRank represented using a matrix N
  - Define N_{ij} to be the share of i's PageRank that j should get in one update step
    - $N_{ij} = 0$ if i doesn't link to j
    - Else N_{ij} is the reciprocal of the number of nodes that i points to
    - **Nodes without outgoing links (~ dangling nodes) pose serious problems to PageRank!**

      Possible solutions:

      - Remove these nodes,
      - For these nodes, define $N_{ii} = 1$,
      - For these nodes, add a complete set of outgoing links to all the other nodes: N_{ij} = 1/n

$$r_i \coloneqq N_{1i}r_1 + N_{2i}r_2 + \cdots + N_{ni}r_n$$
$$r \coloneqq N^T r$$

Link Analysis: PageRank

Spectral Analysis of PageRank

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  - $N_{ij} = 0$ if i doesn't link to j
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    Possible solutions:
    - Remove these nodes,
    - For these nodes, define $N_{ii} = 1$,
    - For these nodes, add a complete set of outgoing links to all the other nodes: $N_{ij} = 1/n$

$$r_i := N_{1i}r_1 + N_{2i}r_2 + \cdots + N_{ni}r_n$$
$$r := N^T r$$

- Scaled PageRank
  - $\tilde{N}_{ij}$: $\tilde{N}_{ij} := sN_{ij} + \frac{1-s}{n}$ and $r := \tilde{N}^T r$

Link Analysis: PageRank

- In general, for matrices such as $\tilde{N}$ that are not symmetric, the eigenvalues can be complex numbers, and the eigenvectors may have less clean relationships to one another
- Fortunately, all entries are positive (i.e., $\tilde{N}_{ij} > 0$ for all entries $\tilde{N}_{ij}$), we can use a powerful result known as *Perron's Theorem*:
  1. $\tilde{N}$ has a real eigenvalue $c > 0$ such that $c > |c'|$ for all other eigenvalues c'.
  2. There is an eigenvector y with positive real coordinates corresponding to the largest eigenvalue c, and y is unique up to multiplication by a constant.
  3. If the largest eigenvalue c is equal to 1, then for any starting vector $x_0 \neq 0$ with nonnegative coordinates, the sequence of vectors $x_{i+1} = \tilde{N}^k x_i$ converges to a vector in the direction of y as $k \to \infty$

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PageRank - Revisited

- The year 1998 was an eventful year for Web link analysis models. Both the PageRank and HITS algorithms were reported in that year.
- The connections between PageRank and HITS are quite striking.
- Since that eventful year, PageRank has emerged as the dominant link analysis model,
  - due to its query-independence,
  - its ability to combat spamming, and
  - Google's huge business success.

Rank Prestige

- In the prestige measures, an important factor is considered,
  - the *prominence* of individual actors who do the “voting”
- In the real world, a person i chosen by an important person is more prestigious than chosen by a less important person.
  - For example, if a company CEO votes for a person is much more important than a worker votes for the person.
- If one’s circle of influence is full of prestigious actors, then one’s own prestige is also high.
  - Thus, one’s prestige is affected by the ranks or statuses of the involved actors.

Rank Prestige (continued)

- Based on this intuition, the rank prestige $P_R(i)$ is defined as a linear combination of ranks of the links that point to i:

$$P_R(i) = A_{1i}P_R(1) + A_{2i}P_R(2) + \cdots + A_{ni}P_R(n)$$

where $A_{ji} = 1$ if j points to i, and 0 otherwise.

  - an actor's rank prestige is a function of the ranks of the actors who vote for the actor,
  - since we have n equations for n actors, we can write them in the matrix notation using $\boldsymbol{P}$ for the vector of all the rank prestige values, i.e., $\boldsymbol{P} = \left(P_R(1), P_R(2), \ldots, P_R(n)\right)^{\mathrm{T}}$
  - the matrix $\boldsymbol{A}$ (where $A_{ij} = 1$ if i points to j, and 0 otherwise) represents the adjacency matrix of the network or graph. We then have: $\boldsymbol{P} = \boldsymbol{A}^{\mathrm{T}}\,\boldsymbol{P}$.
  - the above equation is the characteristic equation used for finding the *eigensystem* of the matrix $\boldsymbol{A}^{\mathrm{T}}$. $\boldsymbol{P}$ is an *eigenvector* of $\boldsymbol{A}^{\mathrm{T}}$.

PageRank: The Intuitive Idea

- **PageRank relies on the democratic nature of the Web** by using its vast link structure as an indicator of an individual page's value or quality.
- PageRank interprets a hyperlink from page j to page i as a vote, by page j, for page i.
- However, PageRank looks at more than the sheer number of votes; it also analyzes the page that casts the vote.
  - Votes casted by “important” pages weigh more heavily and help to make other pages more "important."
- This is exactly the idea of *rank prestige* in social network.

More Specifically

- A hyperlink from a page to another page is an implicit conveyance of authority to the target page.
  - The more in-links that a page i receives, the more prestige the page i has.
- Pages that point to page i also have their own prestige scores.
  - A page of a higher prestige pointing to i is more important than a page of a lower prestige pointing to i.
  - In other words, a page is important if it is pointed to by other important pages.

PageRank Algorithm

- According to *rank prestige*, the importance of page i (i's PageRank score) is the sum of the PageRank scores of all pages that point to i.
- Since a page may point to many other pages, its prestige score should be shared.
- The Web as a directed graph $G = (V, E)$. Let the total number of pages be n. The PageRank score of the page i (denoted by $P(i)$) is defined by:

$$P(i) = \sum_{(j,i)\in E} \frac{P(j)}{O_j},$$

O_j is the number of out-links of j

Matrix Notation

- We have a system of n linear equations with n unknowns. We can use a matrix to represent them.
- Let $\boldsymbol{P}$ be a n-dimensional column vector of PageRank values, i.e.,
$$\boldsymbol{P} = \left(P(1), P(2), \dots, P(n)\right)^{\mathrm{T}}.$$
- Let $\boldsymbol{A}$ be the adjacency matrix of our graph with
$$A_{ij} = \begin{cases} \dfrac{1}{O_i} & \text{if } (i,j) \in E \\ 0 & \text{otherwise} \end{cases} \tag{LA1}$$
- We can write the n equations with (*PageRank*):
$$\boldsymbol{P} = \boldsymbol{A}^{\mathrm{T}} \boldsymbol{P} \tag{LA2}$$

Solve the PageRank Equation: $\boldsymbol{P} = \boldsymbol{A}^{\mathrm{T}}\boldsymbol{P}$

- This is the characteristic equation of the *eigensystem*, where the solution to $\boldsymbol{P}$ is an *eigenvector* with the corresponding *eigenvalue* of 1.
- It turns out that if *some conditions* are satisfied, 1 is the largest *eigenvalue* and the PageRank vector $\boldsymbol{P}$ is the *principal eigenvector*.
- A well-known mathematical technique called ***power iteration*** can be used to find $\boldsymbol{P}$.
- **Problem**: the above Equation does not quite suffice because the Web graph does not meet the conditions.

Using Markov Chain

- *To introduce these conditions* and the enhanced equation, let us derive the same Equation (LA2) based on the **Markov chain**.
  - In the Markov chain, each Web page or node in the Web graph is regarded as a state.
  - A hyperlink is a transition, which leads from one state to another state with a probability.
- This framework models Web surfing as a stochastic process.
- It *models a Web surfer* randomly surfing the Web as state transition.

Random Surfing

- Recall we use O_i to denote the number of out-links of a node i.
- Each transition probability is $1/O_i$ if we assume the Web surfer will click the hyperlinks in the page i uniformly at random.
  - The “back” button on the browser is not used and
  - the surfer does not type in an URL.

Transition Probability Matrix

- Let ***A*** be the state transition probability matrix,

$$A = \begin{pmatrix} A_{11} & A_{12} & . & . & . & A_{1n} \\ A_{21} & A_{22} & . & . & . & A_{2n} \\ . & . & & & & . \\ . & . & & & & . \\ . & . & & & & . \\ A_{n1} & A_{n2} & . & . & . & A_{nn} \end{pmatrix}$$

- A_{ij} represents the transition probability that the surfer in state i (page i) will move to state j (page j). A_{ij} is defined exactly as in Equation (LA2).

Let Us Start

- Given an initial probability distribution vector that a surfer is at each state (or page)
  - $\boldsymbol{p_0} = \left(p_0(1), p_0(2), \dots, p_0(n)\right)^{\mathrm{T}}$ (a column vector) and
  - an $n \times n$ transition probability matrix $\boldsymbol{A}$,

we have: $\sum_{i=1}^{n} p_0(i) = 1\,, \qquad \sum_{j=1}^{n} A_{ij} = 1 \qquad \text{(LA3)}$

- If the matrix $\boldsymbol{A}$ satisfies Equation (LA3), we say that $\boldsymbol{A}$ is the stochastic matrix of a Markov chain.

Back to the Markov Chain

- In a Markov chain, a question of common interest is:
  - Given $\boldsymbol{p}_0$ at the beginning, what is the probability that m steps/transitions later the Markov chain will be at each state j?
- We determine the probability that the system (or *the random surfer*) is in state j after 1 step (1 transition) by using the following reasoning:

$$p_1(j) = \sum_{i=1}^{n} A_{ij}(1) p_0(i)$$

State Transition

$$p_1(j) = \sum_{i=1}^{n} A_{ij}(1) p_0(i)$$

where $A_{ij}(1)$ is the probability of going from i to j in 1 transition, and $A_{ij}(1) = A_{ij}$. We can write it in a matrix form:

$$\boldsymbol{p_1} = \boldsymbol{A}^{\mathrm{T}} \boldsymbol{p_0}$$

In general, the probability distribution after k steps/transitions is:

$$\boldsymbol{p_k} = \boldsymbol{A}^{\mathrm{T}} \boldsymbol{p_{k-1}} \,.$$

Stationary Probability Distribution

- By a Theorem of the Markov chain,
  - a finite Markov chain defined by the ***stochastic matrix*** $\boldsymbol{A}$ has a unique ***stationary probability distribution*** if $\boldsymbol{A}$ is ***irreducible*** and ***aperiodic***.
- The stationary probability distribution means that after a series of transitions $\boldsymbol{p}_k$ will converge to a steady-state probability vector π regardless of the choice of the initial probability vector $\boldsymbol{p}_0$, i.e.,

$$\lim_{k \to \infty} \boldsymbol{p}_k = \boldsymbol{\pi}$$

PageRank Again

- When we reach the steady-state, we have $\boldsymbol{p}_k = \boldsymbol{p}_{k+1} = \pi$, and thus

$$\pi = \boldsymbol{A}^{\mathrm{T}}\pi.$$

- π is the ***principal eigenvector*** of $\boldsymbol{A}^{T}$ with ***eigenvalue*** of 1.
- In PageRank, π *is used as the PageRank vector* $\boldsymbol{P}$. We again obtain Equation (LA2), which is reproduced here:

$$\boldsymbol{P} = \boldsymbol{A}^{\mathrm{T}}\boldsymbol{P} \qquad \text{(LA4)}$$

Is $P = \pi$ Justified?

- Using the stationary probability distribution π as the PageRank vector is reasonable and quite intuitive because
  - it reflects the long-run probabilities that a random surfer will visit the pages.
  - A page has a high prestige if the probability of visiting it is high.

Back to the Web Graph

- Now let us come back to the real Web context and see whether the above conditions are satisfied, i.e.,
  - whether $\boldsymbol{A}$ is a *stochastic matrix* and
  - whether it is *irreducible* and *aperiodic*.
- None of them is satisfied!
- Hence, we need to extend the ideal-case Equation (LA4) to produce the “actual PageRank” model.

A is Not a Stochastic Matrix

- $\boldsymbol{A}$ is the transition matrix of the Web graph

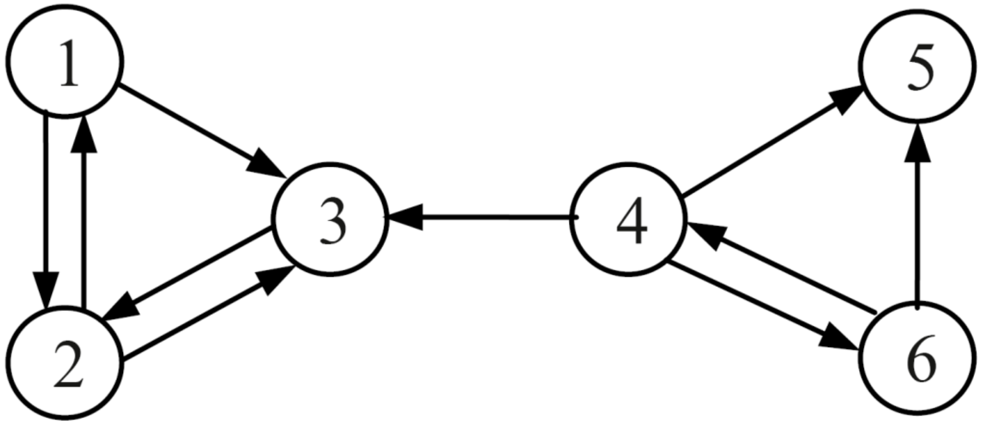
$$A_{ij} = \begin{cases} \dfrac{1}{O_i} & \text{if } (i,j) \in E \\ 0 & \text{otherwise} \end{cases}$$

- It does not satisfy equation (LA3)

$$\sum_{j=1}^{n} A_{ij} = 1$$

- because many Web pages have no out-links, which are reflected in transition matrix $\boldsymbol{A}$ by some rows of complete 0's.
  - Such pages are called the *dangling pages* (nodes).

An Example Web Hyperlink Graph



$$A = \begin{pmatrix} 0 & 1/2 & 1/2 & 0 & 0 & 0 \\ 1/2 & 0 & 1/2 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1/3 & 0 & 1/3 & 1/3 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1/2 & 1/2 & 0 \end{pmatrix}$$

Fix the Problem: Two Possible Ways

1. Remove those pages with no out-links during the PageRank computation as these pages do not affect the ranking of any other page directly.
2. Add a complete set of outgoing links from each such page i to all the pages on the Web.

Let us use the second way

$$\bar{\boldsymbol{A}} = \begin{pmatrix} 0 & 1/2 & 1/2 & 0 & 0 & 0 \\ 1/2 & 0 & 1/2 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1/3 & 0 & 1/3 & 1/3 \\ 1/6 & 1/6 & 1/6 & 1/6 & 1/6 & 1/6 \\ 0 & 0 & 0 & 1/2 & 1/2 & 0 \end{pmatrix}$$

A Is Not Irreducible

- *Irreducible* means that the Web graph G is *strongly connected*.

  Definition: A directed graph $G = (V, E)$ is *strongly connected* if and only if, for each pair of nodes $u, v \in V$, there is a path from u to v.

- A general Web graph represented by $\boldsymbol{A}$ is not irreducible because
  - for some pair of nodes u and v, there is no path from u to v.
  - In our example, there is no directed path from node 3 to 4.

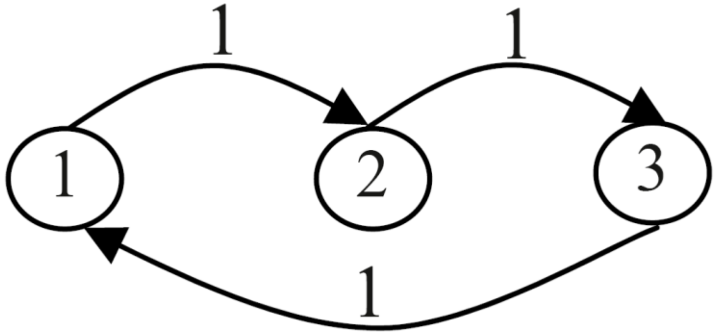
A Is Not Aperiodic

- A state i in a Markov chain being periodic means that there exists a directed cycle that the chain has to traverse.
- **Definition:** A state i is *periodic* with period $k > 1$ if k is the smallest number such that all paths leading from state i back to state i have a length that is a multiple of k.
  - If *a state* is not periodic (i.e., $k = 1$), it *is* *aperiodic*.
  - *A Markov chain is* *aperiodic* if all states are aperiodic.

An Example: Periodic

Figure below shows a periodic Markov chain with $k = 3$. E.g., if we begin from state 1, to come back to state 1 the only path is $1 - 2 - 3 - 1$ for some number of times, say h. Thus, any return to state 1 will take $3h$ transitions.

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$



A periodic Markov chain with $k = 3$.

Deal with Irreducible and Aperiodic

- It is easy to deal with the above two problems with a single strategy.
- Add a link from each page to every page and give each link a small transition probability controlled by a parameter d.
- Obviously, the augmented transition matrix becomes *irreducible* and *aperiodic*

Improved PageRank

- After this augmentation, at a page, the random surfer has two options
  - With probability d, he randomly chooses an out-link to follow.
  - With probability $1 - d$, he jumps to a random page
- The following equation gives the improved model

$$\boldsymbol{P} = \left((1 - d)\frac{\boldsymbol{E}}{n} + d\boldsymbol{A}^{\mathrm{T}} \right) \boldsymbol{P} \qquad \text{(LA5)}$$

where $\boldsymbol{E}$ is $\boldsymbol{e}\boldsymbol{e}^{\mathrm{T}}$ ($\boldsymbol{e}$ is a column vector of all 1's) and thus $\boldsymbol{E}$ is a $n \times n$ square matrix of all 1's.

Follow Our Example

$$(1-d)\frac{\boldsymbol{E}}{n}+d\boldsymbol{A}^{\mathrm{T}}=\begin{pmatrix} 1/60 & 7/15 & 1/60 & 1/60 & 1/6 & 1/60 \\ 7/15 & 1/60 & 11/12 & 1/60 & 1/6 & 1/60 \\ 7/15 & 7/15 & 1/60 & 19/60 & 1/6 & 1/60 \\ 1/60 & 1/60 & 1/60 & 1/60 & 1/6 & 7/15 \\ 1/60 & 1/60 & 1/60 & 19/60 & 1/6 & 7/15 \\ 1/60 & 1/60 & 1/60 & 19/60 & 1/6 & 1/60 \end{pmatrix}$$

The Final PageRank Algorithm

- $(1-d)\frac{\boldsymbol{E}}{n} + d\boldsymbol{A}^{\mathrm{T}}$ is a **stochastic matrix** (transposed). It is also **irreducible** and **aperiodic**.
- If we scale Equation (LA5) so that $\boldsymbol{e}^{\mathrm{T}}\boldsymbol{P} = n$,

$$\boldsymbol{P} = (1-d)\boldsymbol{e} + d\boldsymbol{A}^{\mathrm{T}}\boldsymbol{P}$$

- PageRank for each page i is

$$\boldsymbol{P}(i) = (1-d) + d\sum_{j=1}^{n} \boldsymbol{A}_{ji}\boldsymbol{P}(j) \qquad \text{(LA6)}$$

The Final PageRank Algorithm (continued)

- (LA6) is equivalent to the formula given in the PageRank paper

$$\boldsymbol{P}(i) = (1-d) + d \sum_{(j,i)\in E} \frac{\boldsymbol{P}(j)}{O_j}$$

- The parameter d is called the *damping factor* which can be set to between 0 and 1. $d = 0.85$ was used in the PageRank paper.

Compute PageRank

Use the *power iteration* method

PageRank-Iterate(G)

$\boldsymbol{P}_0 \leftarrow \boldsymbol{e}/n$

$k \leftarrow 1$

repeat

$\boldsymbol{P}_{k+1} \leftarrow (1-d)\boldsymbol{e} + d\boldsymbol{A}^{\mathrm{T}}\boldsymbol{P}_k$

$k \leftarrow k+1$

until $\|\boldsymbol{P}_{k+1} - \boldsymbol{P}_k\|_{\mathbf{1}} < \varepsilon$

return $\boldsymbol{P}_{k+1}$

Advantages of PageRank

- **Fighting spam:** A page is important if the pages pointing to it are important.
  - Since it is not easy for Web page owner to add in-links into his/her page from other important pages, it is thus not easy to influence PageRank.
- **PageRank is a global measure and is query independent:**
  - PageRank values of all the pages are computed and saved off-line rather than at the query time.
- **Criticism:** Query-independence. It could not distinguish between pages that are authoritative in general and pages that are authoritative on the query topic.

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- *Themed Communities and Their Discovery*

HITS - Revisited

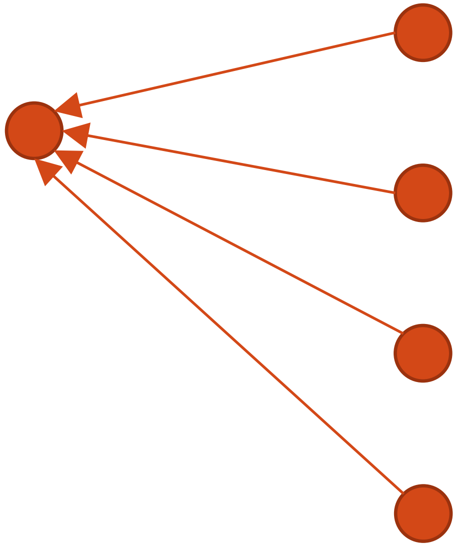
- Unlike PageRank which is a static ranking algorithm, HITS (~ Hypertext Induced Topic Search) is search query dependent.
- When the user issues a search query,
  - HITS first expands the list of relevant pages returned by a search engine and
  - then produces two rankings of the expanded set of pages, *authority ranking* and *hub ranking*.

Authorities and Hubs

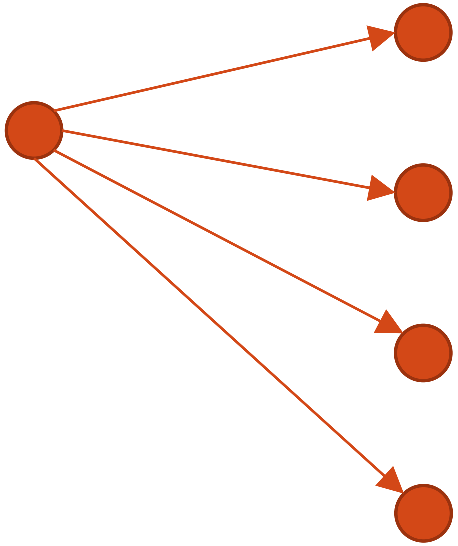
- **Authority:** Roughly, it is a page with many in-links.
  - The idea is that the page may have good or authoritative content on some topic and
  - thus, many people trust it and link to it.
- **Hub:** A hub is a page with many out-links.
  - The page serves as an organizer of the information on a particular topic and
  - points to many good authority pages on the topic.

Examples

An authority

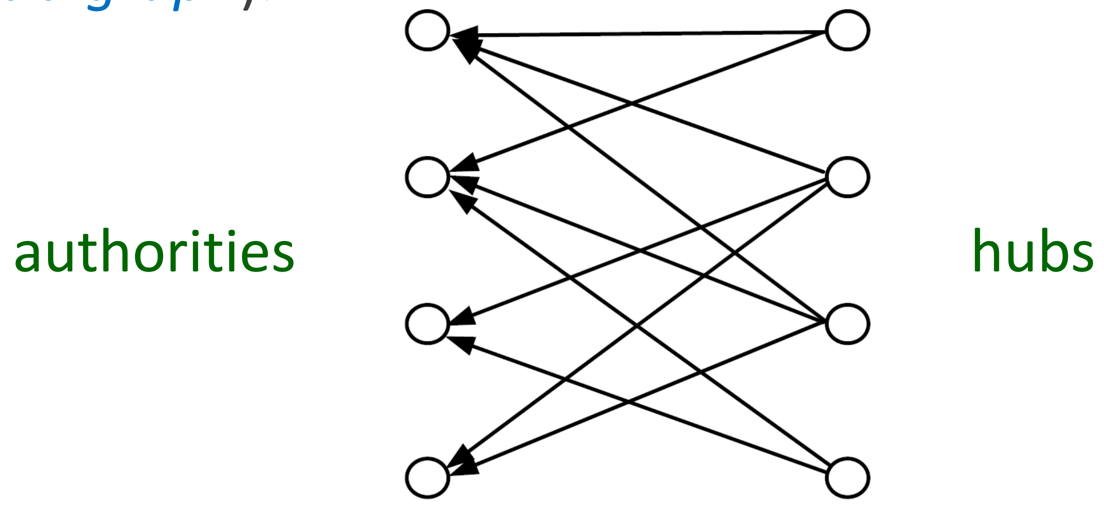


A hub



The key idea of HITS

- A good hub points to many good authorities, and
- A good authority is pointed to by many good hubs.
- Authorities and hubs have a mutual reinforcement relationship. The figure below shows some densely linked authorities and hubs (*a bipartite sub-graph*).



The HITS Algorithm: Grab Pages

- Given a broad search query q, HITS collects a set of pages as follows:
  - It sends the query q to a search engine.
  - It then collects t ($t = 200$ is used in the HITS paper) highest ranked pages. This set is called the *root set* W.
  - It then grows W by including any page pointed to by a page in W and any page that points to a page in W. This gives a larger set S, *the base set*.

The link graph G

- HITS works on the pages in S and assigns every page in S an *authority score* and a *hub score*.
- Let the number of pages in S be n.
- We again use $G = (V, E)$ to denote the hyperlink graph of S.
- We use L to denote the adjacency matrix of the graph.

$$L_{ij} = \begin{cases} 1 & \text{if } (i, j) \in E \\ 0 & \text{otherwise} \end{cases}$$

The HITS Algorithm

- Let the authority score of the page i be $a(i)$, and the hub score of page i be $h(i)$.
- The mutual reinforcing relationship of the two scores is represented as follows:

$$a(i) = \sum_{(j,i)\in E} h(j) \quad \text{and} \quad h(i) = \sum_{(i,j)\in E} a(j)$$

HITS in the Matrix Form

- We use $\boldsymbol{a}$ to denote the column vector with all the authority scores,

$$\boldsymbol{a} = \left(a(1), a(2), \dots, a(n)\right)^{\mathrm{T}}$$, and

- use $\boldsymbol{h}$ to denote the column vector with all the authority scores,

$$\boldsymbol{h} = \left(h(1), h(2), \dots, h(n)\right)^{\mathrm{T}}$$

- Then,

$\boldsymbol{a} = L^{\mathrm{T}}\boldsymbol{h}$ and $\boldsymbol{h} = L\boldsymbol{a}$

Computation of HITS

- The computation of authority scores and hub scores is the same as the computation of the PageRank scores, using the so-called *power iteration*.
- If we use $\boldsymbol{a}_k$ and $\boldsymbol{h}_k$ to denote authority and hub vectors at the k-th iteration, the iterations for generating the final solutions are

$$\boldsymbol{a}_k = L^{\mathrm{T}} L \boldsymbol{a}_{k-1}$$
$$\boldsymbol{h}_k = L L^{\mathrm{T}} \boldsymbol{h}_{k-1}$$

starting with

$$\boldsymbol{a}_0 = \boldsymbol{h}_0 = (1, 1, \dots, 1)^{\mathrm{T}}$$

The Algorithm

HITS-Iterate(G)

$\boldsymbol{a}_0 \leftarrow \boldsymbol{h}_0 \leftarrow (1,1,\dots,1)^{\mathrm{T}}$

$k \leftarrow 1$

repeat

$\boldsymbol{a}_k \leftarrow \boldsymbol{L}^{\mathrm{T}}\boldsymbol{L}\boldsymbol{a}_{k-1}$

$\boldsymbol{h}_k \leftarrow \boldsymbol{L}\boldsymbol{L}^{\mathrm{T}}\,\boldsymbol{h}_{k-1}$

$\boldsymbol{a}_k \leftarrow \boldsymbol{a}_k / \|\boldsymbol{a}_k\|_1$ // normalization

$\boldsymbol{h}_k \leftarrow \boldsymbol{h}_k / \|\boldsymbol{h}_k\|_1$ // normalization

$k \leftarrow k + 1$

until $\|\boldsymbol{a}_k - \boldsymbol{a}_{k-1}\|_1 < \varepsilon_a$ and $\|\boldsymbol{h}_k - \boldsymbol{h}_{k-1}\|_1 < \varepsilon_h$

return $\boldsymbol{a}_k$ and $\boldsymbol{h}_k$

Relationships with Co-Citation and Bibliographic Coupling

- Co-citation of pages i and j is denoted by C_{ij} (e.g., the number of papers that co-cite papers i and j):

$$C_{ij} = \sum_{k=1}^{n} L_{ki} L_{kj} = (L^{\mathrm{T}} L)_{ij}$$

  - the authority matrix $(L^{\mathrm{T}}L)$ of HITS is the *co-citation matrix* C

- Bibliographic coupling of two pages i and j is denoted by B_{ij} (e.g., the number of papers that are cited by both papers i and j):

$$B_{ij} = \sum_{k=1}^{n} L_{ik} L_{jk} = (\mathrm{LL}^{T})_{ij},$$

  - the hub matrix (LL^{T}) of HITS is the *bibliographic coupling matrix* B

Strengths and Weaknesses of HITS

- Strength: its ability to rank pages according to the query topic, which may provide more relevant authority and hub pages.
- Weaknesses:
  - It can be easily spammed: It is in fact quite easy to influence HITS since adding out-links in one's own page is so easy.
  - Topic drift: Many pages in the expanded set may not be on topic.
  - Inefficiency at query time: The query time evaluation is slow. Collecting the root set, expanding it and performing eigenvector computation are all expensive operations

HITS and PageRank – the Summary

- Within the framework of link analysis, we introduced
  - The principles of centrality and prestige
  - Co-citation and bibliographic coupling
  - The algorithms HITS and PageRank, which powers Google
- Yahoo! and MSN have their own link-based algorithms (not published).
- **Important:** Hyperlink based ranking is not the only algorithm used in search engines. In fact, it is combined with many content-based factors to produce the final ranking presented to the user.

Social Networks and their Analysis

Homophily, Social Influence, and Affiliation

- *Homophily Test*
- *Selection and Social Influence*
- *Affiliation Networks*
- *Schelling Model of Segregation*
- *Positive and Negative Relationships*

Link Analysis

- *A Brief Review of Actor-Level and Network-Level Measures*
- *Hubs and Authorities, the HITS Algorithm*
- *The PageRank Algorithm*
- *PageRank – Revisited*
- *HITS – Revisited*
- **Themed Communities and Their Discovery**

Link Analysis and the Discovery of Themed Communities

- Links can also be used to find *themed communities*, which are groups of content-creators or people sharing some common interests.
  - Web communities
  - Email communities
  - Named entity communities
- Focused crawling: combining contents and links to crawl Web pages of a specific topic.
  - Follow links and
  - Use learning/classification to determine whether a page is on topic.

Discovery of Themed Communities: Problem Definition (1)

Given a finite set of **entities** $S = \{s_1, s_2, \dots, s_n\}$ of the same **type**, a **themed community** is a pair $C = (T, G)$, where T is the **community theme** and $G \subseteq S$ is the set of all entities in S that share the theme T. If $s_i \in G$, s_i is said to be a **member** of the community C.

- **A theme defines a community:** given a theme T, the set of members of the community is uniquely determined. Two communities are thus equal if they have the same theme.
- **A theme can be defined arbitrarily:** it can be, e.g., an event (like a sport event or a scandal) or a concept (like web mining).
- **An entity s_i in S can be in any number of communities:** communities can thus overlap, and multiple communities may share members.
- **The entities in S are of the same type:** e.g., people and organizations thus cannot be in the same community.

Discovery of Themed Communities: Problem Definition (2)

Communities can form a hierarchical structure:

- A themed community (T, G) may include a set of **themed sub-communities** $\{(T_1, G_1), \dots, (T_m, G_m)\}$, where T_i is a **sub-theme** of T and $G_i \subseteq G$.
- (T, G) is also called a **themed super-community** of (T_i, G_i).
- In the same way, each sub-community (T_i, G_i) can be further decomposed, which gives us a **hierarchy of a themed community**.

The objective of themed community discovery:

Given a data set containing entities, we want to discover hidden communities of the entities. For each community, we want to find the theme and its members. The theme is usually represented by a set of keywords.

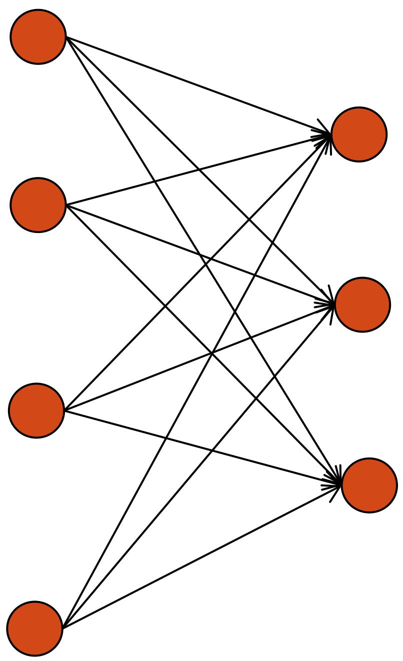
Discovery of Themed Communities: Problem Definition (3)

Community manifestation in the data:

- community members are linked in some way
- the associated texts contain words indicative of the community theme
- **Web pages:**
  1. **Hyperlinks:** the members of a community are more likely to be mutually inter-connected through hyperlinks
  2. **Content words:** community web pages contain words related to the community theme
- **E-mails:**
  1. **E-mail exchange between entities:** members of a community are more likely to communicate with one another
  2. **Content words:** e-mails of community members contain words related to community theme
- **Text documents:**
  1. **Co-occurrence of entities:** community members are more likely to appear together in the same sentence and/or the same document
  2. **Content words:** words in the sentences indicate the community theme

Bipartite Core Communities

A $(4,3)$-bipartite core:



4 fans 3 centers

- an (i,j)-*bipartite core* is a complete bipartite sub-graph with at least i fan nodes and at least j center nodes, i.e., contains all possible edges between the i fan vertices and the j center vertices
- the *fans* represent specialized *hubs*
- the *centers* correspond to *authorities*
- the following algorithm finds only the core pages of the communities, but not all the member pages.
- the algorithm also does not find the themes of the communities nor their hierarchical structure

Bipartite Core Communities

Given a large number of pages crawled from the web, the procedure for finding bipartite cores consists of:

Step 1: **Pruning**

- Pruning by in-degree: delete all the pages that are highly referenced, i.e., their number of in-links is greater than some k (e.g., $k = 50$)
- Iterative pruning of fans and centers (to find the (i, j)-cores):
  - prune all the fans with an out-degree smaller than j
  - prune all the centers with an in-degree smaller than i
  - delete the edges associated with the pruned nodes from the graph

Step 2: **Generating all (i, j)-cores**

- find all $(1, j)$-cores, i.e., the set of all vertices with out-degree at least j
- construct all $(2, j)$-cores by checking every fan which also points to any center in an $(1, j)$-core
- continue in the same fashion up to i, by checking every fan which also points to any center in any of the previously found cores

Overlapping Communities of Named Entities

- **The objective:** **find entity communities from a text corpus**
- **The main idea:**
  - **two entities are linked if they co-occur in the same sentence**
    - e.g., if two people are mentioned in the same sentence, there is usually a relationship between them
  - **a single entity can appear in multiple communities**
  - **the top keywords are assumed to represent the theme of the community**
- **Promising results** when applied for finding communities of political figures and celebrities from web documents

Overlapping Communities of Named Entities: the Algorithm

1. **Build a link graph**
   - parse each document and for each sentence, identify named entities contained in it
   - if a sentence has more than one named entity, these entities are pair-wise linked.
   - The keywords in the sentence are attached to the linked pairs as textual contents.
   - Discard all the other sentences

2. **Find all triangles**
   - A triangle consists of three entities bound together and represents the basic building block of communities
   - observation: a community tends to expand by triangles sharing a common edge

3. **Find community cores**
   - A community core is a group of tightly bound triangles, which are relaxed complete sub-graph (or cliques).
   - Intuitively, a core consists of a set of tightly connected members of a community.

4. **Cluster around community cores**
   - For those triangles and also entity pairs that are not in any core, they are assigned to cores according to their textual content similarities with the discovered cores.

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